

Mind Your Entropy: From Maximum Entropy to Trajectory Entropy-Constrained RL

Guojian Zhan^{1 2 *} Likun Wang^{1 *} Pengcheng Wang² Feihong Zhang¹ Jingliang Duan³ Kaicheng Yu⁴
Masayoshi Tomizuka² Shengbo Eben Li¹

Abstract

Maximum entropy has become a mainstream off-policy reinforcement learning (RL) framework for balancing exploitation and exploration. However, two bottlenecks still limit further performance gains: (1) *non-stationary Q-value estimation* stemming from the joint injection of entropy and the concurrent updating of its temperature parameter; and (2) *short-sighted local entropy tuning*, which adjusts temperature solely based on current single-step entropy without accounting for cumulative entropy over time. In this paper, we broaden the maximum entropy framework by proposing a trajectory entropy-constrained reinforcement learning (TECRL) framework to address these limitations. We begin by introducing reward-entropy separation (RES) to decouple the value targets, ensuring they remain stable and unaffected by temperature fluctuations. Subsequently, the resulting entropy Q-function is leveraged to explicitly quantify expected cumulative entropy, allowing for the enforcement of a trajectory entropy constraint (TEC) to govern long-term stochasticity. We instantiate this framework as DSAC-E, a practical off-policy algorithm that builds upon the latest distributional soft actor-critic. Extensive evaluations across 10 challenging tasks in locomotion, robotic manipulation, and vision-based driving domains demonstrate that DSAC-E consistently outperforms baselines in both cumulative returns and training stability.

1. Introduction

Balancing exploration and exploitation remains a central challenge in reinforcement learning (RL) (Sutton & Barto, 2018; Li, 2023). To address this, off-policy methods have leveraged the maximum entropy principle, which encourages agents to act as randomly as possible while still optimizing for high returns (Wang et al., 2022; Haarnoja et al., 2017). By augmenting the objective with a temperature-weighted entropy term, algorithms like Soft Actor-Critic (SAC) (Haarnoja et al., 2018a) and its distributional variant DSAC (Duan et al., 2021; 2025) have achieved state-of-the-art performance across a wide range of high-dimensional continuous control tasks, underscoring their practical effectiveness and stability (Eysenbach & Levine, 2022).

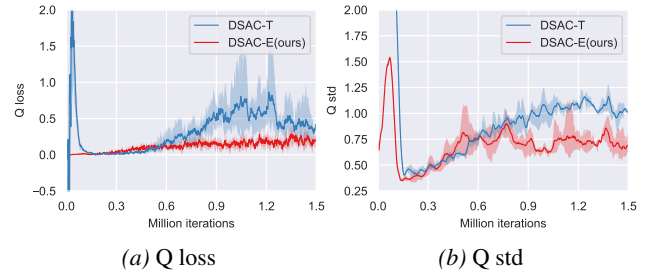


Figure 1. Evidence of addressing (1) non-stationary Q-value estimation. See Appendix B.1 for extended results and analysis.

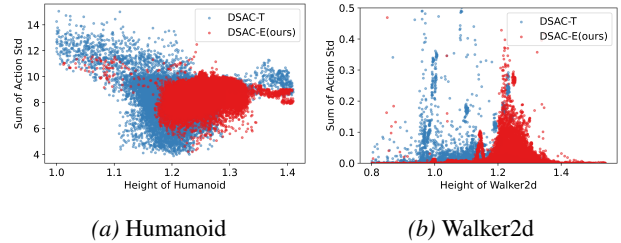


Figure 2. Evidence of addressing (2) short-sighted local entropy tuning. See Appendix B.2 for extended results and analysis.

Determining the entropy temperature is a pivotal aspect of maximum entropy RL, as it directly governs the trade-off between exploration and exploitation. A fixed temperature parameter can lead to a policy that is either excessively

*Equal contribution ¹School of Vehicle and Mobility & College of AI, Tsinghua University ²Berkeley AI Research (BAIR), UC Berkeley ³Westlake University ⁴University of Science and Technology Beijing. Correspondence to: Shengbo Eben Li <lishbo@tsinghua.edu.cn>.

stochastic or prematurely deterministic (Rawlik et al., 2012). This is because a single temperature value cannot optimally balance exploration and exploitation across all phases of training; a high temperature may hinder convergence, while a low temperature can lead to premature exploitation of a suboptimal solution (Fox et al., 2016). To mitigate this issue, modern maximum entropy RL generally incorporates an automated temperature adjustment mechanism (Haarnoja et al., 2018b). Using the policy’s current per-step entropy as a feedback signal, this mechanism dynamically tunes the temperature throughout training, aligning the feedback signal with a predefined target. Therefore, it ensures that a desired level of stochasticity is maintained across all situations (Hazan et al., 2019).

Despite the remarkable empirical success, maximum entropy methods still face two critical bottlenecks that hinder further progress. (1) The first issue is *non-stationary Q-value estimation*, which stems from the tight coupling of reward and entropy (Schulman et al., 2017a). Since the temperature parameter is updated simultaneously, the injected temperature-weighted entropy term is directly altering the Q-value targets, causing them to become non-stationary. This process can destabilize value learning and ultimately undermine policy optimization (Lillicrap et al., 2016; Zhan et al., 2025b). We acknowledge that the bootstrapping update mechanism of Q-values contributes significantly to the non-stationarity in RL. In this context, we highlight that the coupling of reward and entropy is another crucial contributing factor, and our method can effectively address and eliminate this factor. (2) Second, and perhaps more fundamentally, while some works have explored constraining entropy (increase the temperature if the entropy at the current step falls below a target value, and decrease it otherwise), they all suffer from *short-sighted local entropy tuning* (Haarnoja et al., 2018b; Duan et al., 2021; 2025). By regulating only the local current-step entropy, these methods neglect the long-term influence of stochasticity over entire trajectories. More critically, why we say they are short-sighted is that they enforce a uniform entropy target for each state, as if every situation demands the same degree of randomness. This one-size-fits-all assumption is overly restrictive and fails to account for the inherent variability in the dynamics of different states. Consequently, the actor update process is compromised, as it neglects the fact that effective exploration should take into consideration both the underlying system dynamics and the agent’s learning progress (Tokic, 2010; Zhan et al., 2024a; Sun et al., 2022). This fundamental disconnect ignores the varying exploration needs of different situations.

The observed two bottlenecks naturally raise a question: *can we move beyond maximum entropy by directly and cleanly controlling what really matters—the cumulative entropy of the policy?* We argue the answer is yes by introducing a

trajectory entropy-constrained (TEC) RL framework. To ensure a stable and interpretable learning process, we propose to completely decouple reward and entropy signals by maintaining two independent Q-functions. This separation yields stationary Q-value targets, while the dedicated entropy critic facilitates a trajectory-level constraint on the policy’s cumulative entropy. This design inherently transcends traditional single-step restrictions, enabling more effective exploitation alongside precisely controllable exploration.

To demonstrate the practical utility of our theoretical framework, we introduce DSAC-E, a concrete instantiation of TECRL built upon the state-of-the-art Distributional Soft Actor-Critic with Three refinements (DSAC-T) (Duan et al., 2025). DSAC-E materializes the core tenets of TECRL, specifically reward-entropy separation (RES) and trajectory-level constraints (TEC). By leveraging the entropy critic to orchestrate a structured decay of exploration as the policy converges, DSAC-E effectively circumvents the “over-exploration” trap in near-optimal regions, ensuring superior stability and asymptotic performance.

Our contributions are summarized in threefold:

- We identify and analyze two bottlenecks in maximum entropy RL: (1) *non-stationary Q-value estimation* and (2) *short-sighted local entropy tuning*, elucidating how they undermine value estimation accuracy and impede effective policy improvement, respectively.
- We introduce TECRL framework to systematically resolve the identified issues. Specifically, reward-entropy separation (RES) isolates reward and entropy signals into dual critics, ensuring stable value estimation independent of temperature updates. Subsequently, the trajectory entropy constraint (TEC) utilizes the decoupled entropy critic to govern the agent’s long-term exploration profile, thereby overcoming the limitations of short-sighted tuning by enabling a global perspective on the agent’s exploration strategy.
- We introduce DSAC-E algorithm, a practical instantiation of our TECRL framework built on DSAC-T. Extensive experiments demonstrate our superior performance and stability in complex continuous control.

2. Preliminaries

Maximum entropy RL. While standard RL maximizes the expected return, maximum entropy RL (Haarnoja et al., 2017) extends this objective by incorporating a policy entropy term as

$$J_{\pi} = \mathbb{E}_{s_t \sim \rho_{\pi}} \left[\sum_{t=0}^{\infty} \gamma^t [r_t + \alpha \mathcal{H}(\pi(\cdot | s_t))] \right], \quad (1)$$

where $\gamma \in (0, 1)$ is the discount factor, ρ_t is the state visitation distribution, α is the temperature coefficient, and the single-step policy entropy $\mathcal{H}$ is expressed as

$$\mathcal{H}(\pi(\cdot|s_t)) = \mathbb{E}_{a_t \sim \pi(\cdot|s_t)} [-\log \pi(a_t|s_t)]. \quad (2)$$

The optimal policy can be derived through a maximum entropy variant of policy iteration, commonly known as soft policy iteration (Wang et al., 2022). This iterative process alternates between two key stages: (1) soft policy evaluation (PEV) and (2) soft policy improvement (PIM).

In soft PEV, provided a policy π , for a given policy π , we can apply the soft Bellman operator $\mathcal{B}^\pi$ to learn the soft Q-value, as shown by the soft Bellman expectation equation:

$$\mathcal{B}^\pi[Q^{\text{soft}}(s, a)] = r + \gamma \mathbb{E}_\pi[Q^{\text{soft}}(s', a') - \alpha \log \pi(a'|s')], \quad (3)$$

where the definition of soft Q-value is

$$Q^{\text{soft}}(s, a) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_t + \sum_{t=1}^{\infty} \gamma^t \alpha \mathcal{H}(\pi(\cdot|s_t)) \right]. \quad (4)$$

One might ask why we write the reward and entropy signals as two separate summation terms. The reason is to highlight the difference in their starting indices. The reward signal is accumulated from the current time step, with a summation index of $t = 0$, while the policy entropy is accumulated from the next time step, with a summation index of $t = 1$. This difference is evident from the soft Bellman expectation equation in (3): the first term on the right-hand side, r , does not have a corresponding policy entropy term at the same time step. In fact, the missing current entropy $\mathcal{H}(\pi(\cdot|s_0))$ occurs in the subsequent soft PIM step.

In soft policy improvement (PIM), the goal is to find a new policy that outperforms the current policy. This is achieved by directly maximizing an entropy-augmented objective, a process equivalent to:

$$\pi_{\text{new}} = \arg \max_{\pi} \mathbb{E}_{s \sim \rho_\pi, a \sim \pi} [Q^{\text{soft}}(s, a) - \alpha \log \pi(a|s)]. \quad (5)$$

The convergence of soft policy iteration to the optimal maximum entropy policy is a well-established result in the field, as shown by (Haarnoja et al., 2017).

Existing temperature tuning. A key advancement in the latest maximum entropy frameworks is the automatic management of the temperature parameter α . Instead of being a fixed hyperparameter, α is treated as a learnable variable. The objective is to minimize

$$J(\alpha) = \mathbb{E}_{a_t \sim \pi} [-\alpha (\log \pi(a_t|s_t) + \mathcal{H}_0)], \quad (6)$$

where the default value of $\mathcal{H}_0$ is commonly set as $-\dim(\mathcal{A})$, i.e., the minus of the number of action dimensions. This

mechanism achieves a dynamic balance between exploration and exploitation by maintaining the policy’s local entropy close to a predefined target entropy $\mathcal{H}_0$ across all situations (Haarnoja et al., 2018a).

3. Method

3.1. Two Bottlenecks of Maximum Entropy RL

Previously, we briefly introduced two bottlenecks that exist in the current maximum entropy RL framework. Now, combining with specific formulas, we will more formally and mathematically explain their origins and their impact on policy learning.

(1) Non-stationary Q-value estimation. In each soft PEV step, as shown in (3), the target value is calculated by

$$y_{\text{target}} = r(s, a) + \gamma [Q^{\text{soft}}(s', a') + \alpha \mathcal{H}(\pi(\cdot|s'))]. \quad (7)$$

When the temperature α is updated at the same time, the target value distribution shifts dynamically. This entanglement injects additional variance and bias into Q-value estimation, degrading subsequent policy improvement steps that rely on stable value predictions.

(2) Short-sighted local entropy tuning. In each soft PIM step, as shown in (6), the existing temperature tuning mechanism aligns every local single-step entropy to a fixed target by adjusting α to match $\mathbb{E}[-\log \pi(a|s)]$ to some desired value. However, it would be better to adjust the trajectory entropy to control the long-term policy stochasticity, which is defined as:

$$\mathcal{H}_{\text{traj}}(s) = \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t \mathcal{H}(\pi(\cdot|s_t)) \middle| s_0 = s \right]. \quad (8)$$

In summary, while the maximum-entropy framework is a powerful tool for policy learning, its effectiveness is still hindered by the two identified bottlenecks. These limitations motivate us to execute reward-entropy separation to ensure clean and stable value learning and rethink maximum entropy RL from a trajectory-level entropy constraint perspective.

3.2. Trajectory Entropy-Constrained RL

To address the two bottlenecks identified earlier, we propose Trajectory Entropy-Constrained Reinforcement Learning (TECRL). It formulates an explicit equality constraint on the trajectory-level entropy to control the policy stochasticity,

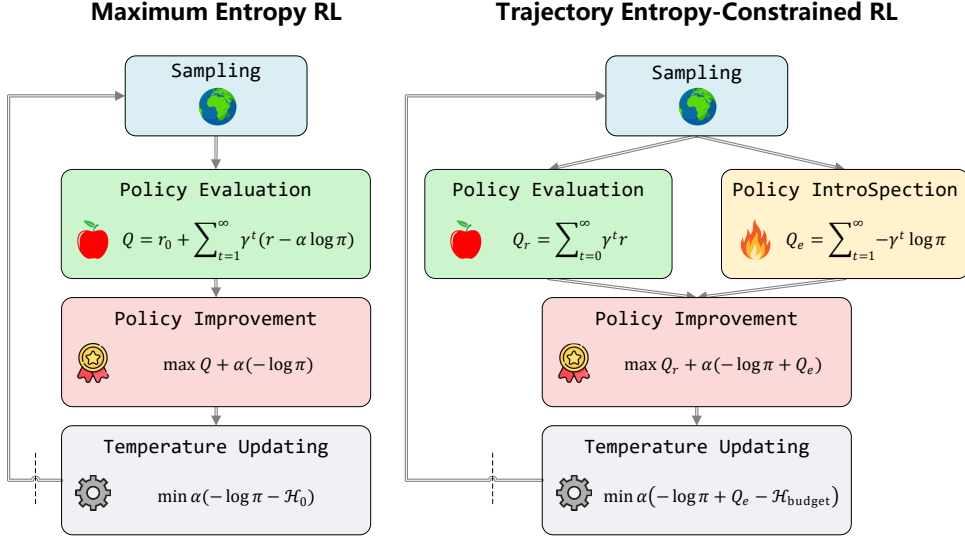


Figure 3. **Comparison** between standard maximum entropy RL (**left**) and our trajectory entropy-constrained (TEC) RL (**right**). Our TECRL framework comprises four key components: a reward-centric policy evaluation (PEV), an entropy-centric policy introspection (PIS), a policy improvement (PIM) that retains the exact soft policy objective, and a temperature update (TUP) tuning the temperature guided by the trajectory entropy constraint.

which yields the following policy optimization problem:

$$\begin{aligned} \max_{\pi} \quad & \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t [r(s_t, a_t)] \right] \\ \text{s.t.} \quad & \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t \mathcal{H}(\pi(\cdot | s_t)) \right] = \mathcal{H}_{\text{budget}}. \end{aligned} \quad (9)$$

Under this trajectory entropy constraint, the agent is required to strategically distribute a fixed budget of randomness across its entire trajectory. This offers a more principled way to mitigate the dilemma of under- and over-exploration.

To practically solve the optimal policy, our TECRL integrates four alternating steps: (1) Policy Evaluation (PEV) estimates the expected cumulative reward; (2) Policy Introspection (PIS) estimates the expected cumulative entropy; (3) Policy Improvement (PIM) jointly leverages both critics to formulate soft policy objective; and (4) Temperature Updating (TUP) adapts the temperature to enforce the trajectory entropy constraint. Below we detail these four steps one by one.

(1) Policy Evaluation (PEV). This step learns a reward-centric critic Q_r defined as

$$Q_r(s, a) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s, a_0 = a \right], \quad (10)$$

Its loss follows the standard Bellman expectation equation:

$$\mathcal{L}_{\text{PEV}} = (Q_r(s, a) - y_r)^2, \quad (11)$$

where $y_r = r(s, a) + \gamma \mathbb{E}_{s', a'} [Q_r(s', a')]$. This reward-centric critic explicitly excludes entropy and the temperature α , ensuring a clean value target uninfluenced by policy stochasticity.

(2) Policy Introspection (PIS). The objective of this step is to learn an entropy-centric critic Q_e . Given that the immediate entropy $\mathcal{H}(\pi(\cdot | s))$ of a Gaussian policy is analytically tractable and can be computed directly, we let Q_e focus exclusively on the expected future uncertainty. Specifically, we define Q_e as the discounted cumulative policy entropy starting from the subsequent time step:

Definition 3.1 (Trajectory-Entropy Critic). The entropy-centric critic $Q_e(s, a)$ is defined as the expected discounted cumulative entropy of the policy trajectories from the next time step to infinity:

$$Q_e(s, a) \triangleq \mathbb{E}_{\pi} \left[\sum_{t=1}^{\infty} \gamma^t \mathcal{H}(\pi(\cdot | s_t)) \mid s_0 = s, a_0 = a \right]. \quad (12)$$

Notably, it also does not contain the temperature α , ensuring its target value is clean and explicit.

To provide a theoretical foundation for learning the entropy-centric critic, we establish that Q_e satisfies a recursive relationship analogous to the standard Bellman equation. This allows us to estimate the long-term trajectory entropy through temporal-difference learning.

Theorem 3.2 (Trajectory Entropy Bellman Equation). *For a fixed policy π , the trajectory-entropy value function $Q_e(s, a)$*

as defined in (12) satisfies the following Bellman expectation equation:

$$Q_e(s, a) = \mathbb{E}_{a' \sim \pi} [\gamma \mathcal{H}(\pi(\cdot|s')) + \gamma Q_e(s', a')]. \quad (13)$$

Proof. See Appendix A.2. $\square$

Based on Theorem 3.2, we optimize Q_e by minimizing the mean-squared Bellman error:

$$\mathcal{L}_{\text{PIS}} = (Q_e(s, a) - y_e)^2, \quad (14)$$

where $y_e = \gamma \mathcal{H}(\pi(\cdot|s')) + \gamma Q_e(s', a')$. We refer to this process as policy introspection because Q_e reflects the future cumulative entropy of the current policy across different state-action pairs. In essence, it quantifies the long-term stochasticity inherent to the policy itself.

(3) Policy Improvement (PIM). With dual critics Q_r and Q_e , We can formulate a policy loss as:

$$\mathcal{L}_{\text{PIM}} = \underbrace{Q_r(s, a)}_{\text{cumulative reward}} + \alpha \underbrace{(-\log \pi(a|s) + Q_e(s, a))}_{\text{cumulative entropy}}. \quad (15)$$

This PIM loss aligns with the soft policy objective shown in (1). Q_r represents the cumulative reward, $-\log \pi(a|s)$ is the current policy entropy, and Q_e represents the cumulative entropy starting from the next time step. Therefore, our PIM is compliant with the maximization term in (9).

(4) Temperature Updating (TUP). Finally, the aim of TUP is tuning α to enforce the trajectory entropy constraint, whose loss is

$$\mathcal{L}_{\text{TUP}} = -\alpha \left(\underbrace{-\log \pi(a|s) + Q_e(s, a)}_{\text{cumulative entropy}} - \mathcal{H}_{\text{budget}} \right). \quad (16)$$

This mechanism extends existing temperature tuning in (6) by replacing uniform local entropy matching with a trajectory-level entropy constraint in (9). We set $\mathcal{H}_{\text{budget}}$ as $\rho H_0 / (1 - \gamma)$. The division by $(1 - \gamma)$ is to keep the magnitude consistent with the local entropy tuning of the maximum entropy. ρ is a scaling factor that can adjust the budget value and we call it budget coefficient.

Summary. Our TECRL framework is grounded in two primary claims: (1) TECRL enables *more stable and effective exploitation*. This stems from the fact that Q^r is decoupled from entropy and temperature terms, ensuring a more accurate and dedicated return prediction for policy improvement. (2) TECRL enables *more strategic and controllable exploration*. By dynamically allocating a finite entropy budget to states where exploration is most critical, the framework facilitates the preservation of high-value behaviors while mitigating erratic fluctuations in policy stochasticity. The pseudocode is summarized in Algorithm 1.

Algorithm 1 TECRL

- 1: **Initialize** policy π_θ , reward critic $Q_{r,\psi}$, entropy critic $Q_{e,\phi}$, temperature α , replay buffer $\mathcal{D}$
 - 2: **for** each iteration **do**
 - 3: Observe s_t , sample $a_t \sim \pi_\theta(a|s_t)$, execute a_t , receive r_t , next state s_{t+1}
 - 4: Store (s_t, a_t, r_t, s_{t+1}) in $\mathcal{D}$
 - 5: Sample mini-batch $\{(s, a, r, s')\} \sim \mathcal{D}$
 - 6: Update Q_r with (11) $\triangleright$ PEV
 - 7: Update Q_e with (14) $\triangleright$ PIS
 - 8: Update π_θ with (15) $\triangleright$ PIM
 - 9: Update α with (16) $\triangleright$ TUP
 - 10: **end for**
-

3.3. Theoretical Analysis

Convergence property. Overall, our TECRL follows the principle of policy iteration. The core of its convergence analysis lies in proving the convergence of Policy Introspection (PIS) step by the following theorem.

Theorem 3.3 (Convergence of TECRL). *The trajectory entropy Q -function Q^e satisfies the contraction mapping condition, there exists a unique solution that can be solved using fixed-point iteration methods.*

Proof. See Appendix A.3. $\square$

Leveraging the contraction mapping principle, we ensure Q_e converges to a unique fixed point, providing a stable foundation for policy training.

Closed-form of optimal policy. In essence, the TEC acts as a reweighting mechanism, where the policy is refined by a weight function $w(s, a) \propto \exp(Q_e(s, a))$ applied to the original reward-centric Boltzmann distribution.

Theorem 3.4 (Optimal Policy Form of TECRL). *For the TECRL formulation in (9), the optimal policy $\pi^*(a|s)$ admits the following closed-form characterization:*

$$\pi^*(a | s) \propto \underbrace{\exp\left(\frac{1}{\alpha} Q_r(s, a)\right)}_{\text{reward Boltzmann}} \cdot \underbrace{w(s, a)}_{\text{TEC}}, \quad (17)$$

where $w(s, a) \propto \exp(Q_e(s, a))$.

Proof. See Appendix A.4 $\square$

This theorem establishes that the optimal policy under TEC is modulated by an entropy-augmented exponential weight. Rather than enforcing uniform exploration, TECRL leverages Q_e to strategically reallocate the exploration budget across states with diverse uncertainty levels.

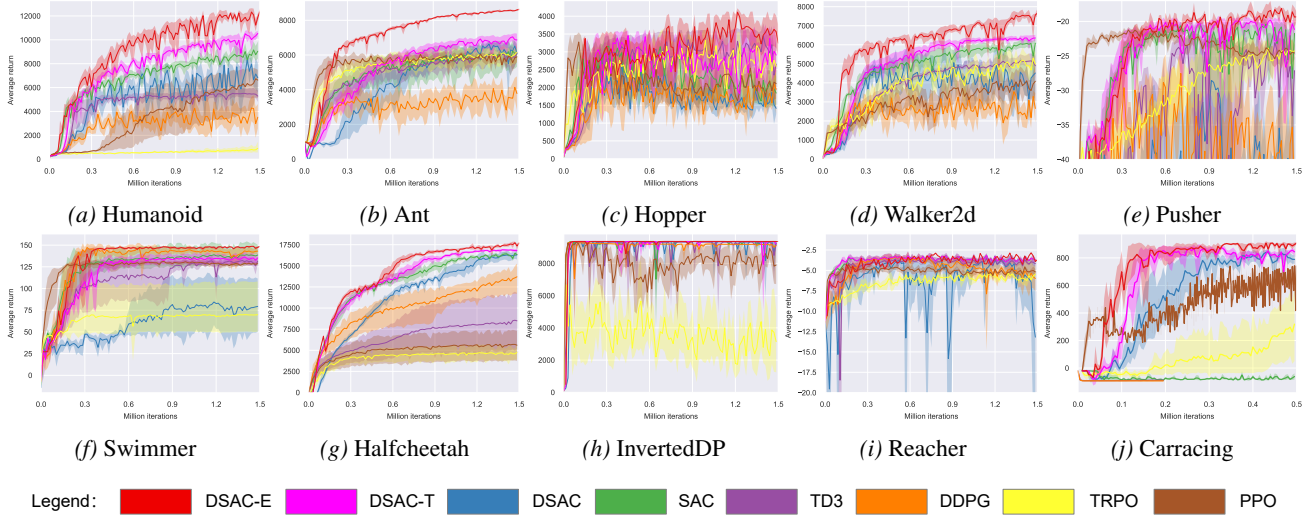


Figure 4. **Training curves.** The solid lines correspond to mean and shaded regions correspond to the confidence interval over five runs.

Table 1. **Average final return.** Computed as the mean of the highest return values observed in the final 10% of iteration steps per run. $\pm$ corresponds to standard deviation over 5 runs.

Algorithm			Humanoid	Ant	Hopper	Walker2d	Pusher
Off-policy	w/ entropy	DSAC-E	12542 ± 280	8640 ± 57	3901 ± 385	7780 ± 137	-17.7 ± 0.2
		DSAC-T	10829 ± 243	7086 ± 261	3660 ± 533	6424 ± 147	-19.1 ± 0.5
		DSAC	9074 ± 286	6862 ± 53	2135 ± 434	5413 ± 865	-25.6 ± 1.3
		SAC	9336 ± 696	6427 ± 805	2483 ± 943	6201 ± 264	-19.6 ± 0.3
	w/o entropy	TD3	5632 ± 436	6184 ± 487	3569 ± 455	5238 ± 336	-21.4 ± 1.2
		DDPG	5292 ± 663	4549 ± 789	2644 ± 659	4096 ± 68	-30.5 ± 6.1
On-policy	w/ entropy	TRPO	965 ± 555	6203 ± 579	3474 ± 400	5503 ± 593	23.0 ± 2.2
		PPO	6869 ± 1563	6157 ± 185	2647 ± 482	4832 ± 638	-22.9 ± 1.4
		↑	15.82%	21.93%	6.58%	21.11%	7.33%

Algorithm			Swimmer	Halfcheetah	InvertedDP	Reacher	Carracing
Off-policy	w/ entropy	DSAC-E	149.3 ± 0.3	17904 ± 100	9360 ± 0	-2.9 ± 0.1	918 ± 3
		DSAC-T	137.6 ± 6.4	17025 ± 157	9360 ± 0	-3.1 ± 0.2	903 ± 7
		DSAC	83.9 ± 35.6	16542 ± 514	9359 ± 1	-4.3 ± 1.9	890 ± 12
		SAC	140.4 ± 14.3	16573 ± 224	9360 ± 0	-3.1 ± 0.2	-12 ± 8
	w/o entropy	TD3	134.0 ± 5.4	8633 ± 4041	9347 ± 15	-3.4 ± 0.2	-89 ± 5
		DDPG	145.6 ± 4.3	13970 ± 2083	9183 ± 10	-4.5 ± 1.3	-93 ± 0
On-policy	w/ entropy	TRPO	70.4 ± 38.1	4785 ± 968	6260 ± 2066	-5.0 ± 0.6	363 ± 323
		PPO	130.3 ± 2.0	5790 ± 2201	9357 ± 2	-4.4 ± 0.2	776 ± 98
		↑	2.54%	5.16%	0%	6.45%	1.66%

* **Bolded** and red = best, blue = second-best. $\uparrow$ means the improvement of the best over the second-best.

Performance bound. We now formalize how TEC influences the policy’s return performance, as shown below.

Theorem 3.5 (Performance Bound of TECRL). *Let π_{soft}^* be the optimal policy under standard maximum entropy RL with optimal return R_{MaxEnt}^* , cumulative entropy $\mathcal{H}_{\text{soft}}^*$, and temperature α_{soft}^* . Let R_{TEC}^* denote the return upper bound for the TECRL policy under an entropy budget $\mathcal{H}_{\text{budget}}$. For any $\mathcal{H}_{\text{budget}} \leq \mathcal{H}_{\text{soft}}^*$, the potential return advantage $\Delta =$*

$R_{\text{TEC}}^* - R_{\text{MaxEnt}}^*$ is bounded by:

$$\Delta \leq \alpha_{\text{soft}}^* (\mathcal{H}_{\text{soft}}^* - \mathcal{H}_{\text{budget}}). \quad (18)$$

Proof. See Appendix A.5. $\square$

Theorem 3.5 establishes a direct correspondence between the performance bound and the entropy gap, $\mathcal{H}_{\text{soft}}^* - \mathcal{H}_{\text{budget}}$. On the basis, a conservative budget facilitates a strategic

trade-off, channeling surplus stochasticity into deterministic returns. Consequently, elevating the performance ceiling becomes not merely an advantage but a necessity; it expands the optimization landscape, allowing the policy to secure superior deterministic gains that were previously unattainable.

4. Experiments

4.1. Main Experiment

Benchmark. We choose 10 challenging continuous control tasks spanning three distinct domains: locomotion, manipulation, and vision-based driving. Comprehensive details regarding the experimental configurations and snapshots are provided in Appendix D.

Baselines. We consider 7 well-known model-free algorithms, including trust region policy optimization (TRPO) (Schulman et al., 2015), proximal policy optimization (PPO) (Schulman et al., 2017b), deep deterministic policy gradient (DDPG) (Lillicrap et al., 2016), twin delayed deep deterministic policy gradient (TD3) (Fujimoto et al., 2018), soft actor-critic (SAC) (Haarnoja et al., 2018a), Distributional SAC (DSAC) (Duan et al., 2021) and its latest version DSAC-T (Duan et al., 2025). See Appendix F for detailed hyperparameters.

Main results. Figure 4 and Table 1 display all the learning curves and numerical performance results, respectively. Our comprehensive findings reveal that across all evaluated 10 tasks, the DSAC-E algorithm consistently matched or surpassed the performance of all competing benchmark algorithms, establishing new state-of-the-art results. Notably, it achieved less oscillation and substantial performance improvements on the Humanoid, Ant, Walker2d, and Pusher tasks, with improvements of 15.82%, 21.93%, 21.11% and 7.3% over the second-best method.

The superior performance on high-dimensional tasks like *Humanoid* can be attributed to the strategic exploration assignment by TEC in DSAC-E. In addition, by decoupling the entropy from the reward by RES, DSAC-E mitigates the destructive interference between the mission-critical reward and the auxiliary exploration, which is often a primary source of instability in standard maximum entropy RL.

4.2. Ablation Study

We conduct two series of ablation studies to evaluate the contribution of each component.

Impact of Reward-entropy separation (RES) and trajectory Entropy Constraint (TEC). We perform a step-wise ablation, considering four algorithms: (1) Our full DSAC-E. (2) DSAC-E w/o TEC, which replaces our trajectory entropy

constraint with existing local entropy tuning. (3) DSAC-E w/o TEC and RES, which is close to DSAC-T but with a ρ -controlled entropy budget. (4) standard DSAC-T.

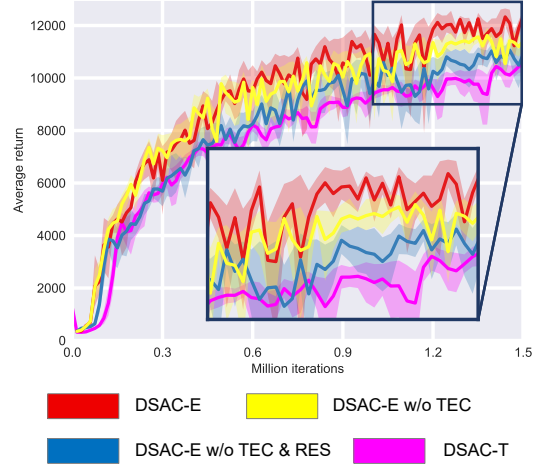


Figure 5. Ablation on TEC and RES using Humanoid task.

Table 2. Results of ablation on TEC and RES.

Algorithm	TAR
DSAC-E (full)	12542 $\pm$ 280
DSAC-E w/o TEC	11786 $\pm$ 374
DSAC-E w/o TEC & RES	11455 $\pm$ 404
DSAC-T	10829 $\pm$ 243

As shown in Figure 5 and Table 2, the performance of the algorithms progressively declines as more components are removed. This result confirms the effectiveness of both our proposed RES and TEC modules.

Impact of the budget coefficient ρ . To evaluate the sensitivity of the hyperparameter ρ , we conducted ablation studies across two representative tasks, Humanoid and Pusher. The learning curves are depicted in Figure 6, and the final performance metrics are summarized in Table 3.

Table 3. Results across different values of ρ .

Hyperparameter	Humanoid	Pusher
$\rho = 1$	11382 $\pm$ 447	-17.7 $\pm$ 0.2
$\rho = 10$	12118 $\pm$ 505	-17.6 $\pm$ 0.4
$\rho = 20$	12542 $\pm$ 280	-17.6 $\pm$ 0.3
$\rho = 30$	11747 $\pm$ 365	-18.0 $\pm$ 0.5

The results reveals that performance remains remarkably stable or exhibits improvement across a broad spectrum of ρ values, suggesting that the algorithm is insensitive to fine-grained parameterization. Specifically, for high-dimensional and unstable locomotion tasks such as *Humanoid*, a moderate increase in ρ (e.g., 10–20) yields substantial performance gains; we observe that a progressive tuning strategy,

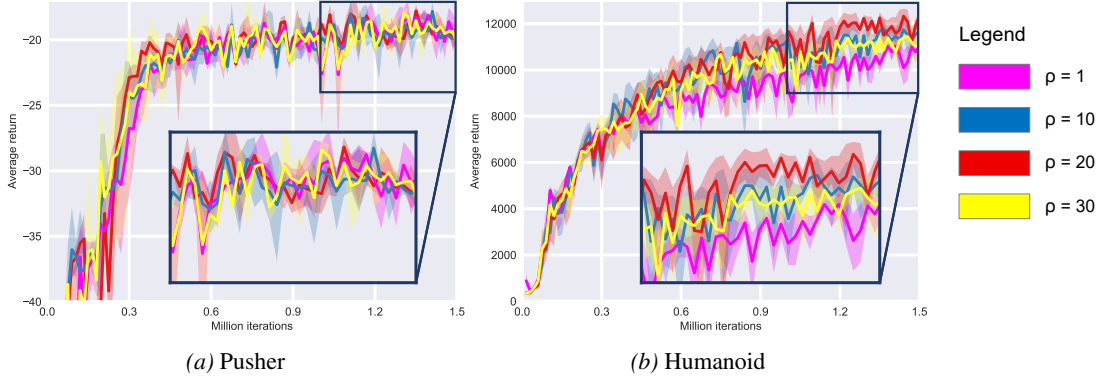


Figure 6. Ablation on the sensitivity to the trajectory entropy budget.

starting from $\rho = 1$ and incrementally increasing the value, consistently facilitates higher returns. Conversely, in relatively lower-dimensional tasks like *Pusher*, the impact of ρ is marginal, with performance remaining consistently near-optimal. This lack of sensitivity in simpler domains indicates that adjusting ρ is unlikely to introduce deleterious side effects. Consequently, we conclude that ρ is highly robust and easily tunable across various task complexities, preserving the core qualitative advantages of TECRL without requiring extensive search.

5. Related Work

Exploration remains a central challenge in RL, and prior studies have proposed various strategies to inject and regulate stochasticity into the policy (Amin et al., 2021). Broadly, existing approaches can be grouped into two main categories: external-noise driven and internal-entropy driven exploration (Hao et al., 2023). While other alternatives, such as curiosity-driven (Sun et al., 2022) or specially designed random distribution-based (An et al., 2021) exploration, have been explored, they remain less commonly adopted in standard model-free RL algorithms.

External-noise driven exploration. A line of methods in off-policy RL encourages exploration by directly perturbing the agent’s actions with a noise process. For instance, DDPG first (Lillicrap et al., 2016) employs Ornstein–Uhlenbeck noise to facilitate temporally correlated exploration, and the TD3 family (Fujimoto et al., 2018; 2023; Seo et al., 2025) turn to simply apply Gaussian noise to each action dimension to effectively maintain randomness during training. Although these approaches are intuitive and easy to implement, they suffer from two key drawbacks. First, the noise is added externally and is entirely separate from the policy’s learning objective. The policy itself is unaware of this exploration mechanism, making it a blind, ad hoc process (Plappert et al., 2018; Li et al., 2021; Wang et al., 2026; Zhan et al., 2025a). Second, it creates a fundamental incon-

sistency between training and evaluation. A policy trained with exploratory noise is different from the final policy used for deployment, which can lead to a policy-value mismatch and hinder convergence to a truly optimal solution (Hollenstein et al., 2022; Sikchi et al., 2022; Zhan et al., 2023). Overall, although action-noise driven exploration is straightforward to implement and can yield good performance, its largely heuristic nature diminishes its reliability.

Internal-entropy driven exploration. A more principled framework for exploration is provided by maximum-entropy RL (Haarnoja et al., 2017). By augmenting the standard RL objective with an internal entropy term, methods such as SAC (Haarnoja et al., 2018a) optimize for both expected return and policy entropy, thereby encouraging diverse behaviors (Nachum et al., 2017; Zhan et al., 2024b). While the latest extensions of SAC further incorporate distributional critics to improve performance (Duan et al., 2021; 2025), they share the same tuning principle of maintaining the policy’s single-step entropy at a fixed target. Recent work has explored the use of generative models, such as diffusion models, as policy functions (Yang et al., 2023b; Zhu et al., 2023; Zhan et al., 2026). While it’s difficult to accurately compute the entropy of this class of functions (Yang et al., 2023a), these methods still try to follow the standard maximum-entropy principle and entropy tuning mechanism for exploration, for example, by approximating the policy entropy via gaussian mixture model fitting or alternatively optimizing the lower bound (Wang et al., 2024; 2025; Ding et al., 2024; Celik et al., 2025). Their entropy tuning mechanism remains inherently uniform across all situations and does not explicitly account for long-term policy stochasticity and the inherent need for adaptive exploration. Our TECRL also employs entropy to monitor policy’s stochasticity. However, we shift the focus from local entropy tuning to *trajectory entropy constraint*, highlighting a new perspective on managing policy’s long-term stochasticity. We believe this work provides a new avenue for better resolving the exploitation-exploration dilemma, leading to higher

performance.

6. Conclusion

In this paper, we propose the trajectory entropy-constrained reinforcement learning (TECRL) framework. By providing a more stable optimization landscape and a long-term perspective on exploration, our framework addresses two fundamental limitations of maximum entropy RL: (1) *non-stationary Q -value estimation* and (2) *short-sighted local entropy tuning*. We instantiate this framework as DSAC-E algorithm. Across 10 rigorous tasks in locomotion, manipulation, and vision-based driving, DSAC-E consistently delivers higher returns and superior stability compared to state-of-the-art baselines. TECRL represents a paradigm shift from point-wise entropy maximization to trajectory-level uncertainty management, offering a robust and scalable solution for complex, continuous control challenges.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

Acknowledgements

This study is supported by the Tsinghua University-Toyota Joint Research Center for AI Technology of Automated Vehicle, Beijing Natural Science Foundation (L257002), and SunRisingAI Lab.

References

- Amin, S., Gomrokchi, M., Satija, H., Van Hoof, H., and Precup, D. A survey of exploration methods in reinforcement learning. *arXiv preprint arXiv:2109.00157*, 2021.
- An, G., Moon, S., Kim, J.-H., and Song, H. O. Uncertainty-based offline reinforcement learning with diversified q-ensemble. *Advances in neural information processing systems*, 34:7436–7447, 2021.
- Brockman, G., Cheung, V., Pettersson, L., Schneider, J., Schulman, J., Tang, J., and Zaremba, W. Openai gym. *arXiv preprint arXiv:1606.01540*, 2016.
- Celik, O., Li, Z., Blessing, D., Li, G., Palenicek, D., Peters, J., Chalkatzaki, G., and Neumann, G. Dime: Diffusion-based maximum entropy reinforcement learning. *arXiv preprint arXiv:2502.02316*, 2025.
- Ding, S., Hu, K., Zhang, Z., Ren, K., Zhang, W., Yu, J., Wang, J., and Shi, Y. Diffusion-based reinforcement learning via q-weighted variational policy optimization. *Advances in Neural Information Processing Systems*, 37:53945–53968, 2024.
- Duan, J., Guan, Y., Li, S. E., Ren, Y., Sun, Q., and Cheng, B. Distributional soft actor-critic: Off-policy reinforcement learning for addressing value estimation errors. *IEEE transactions on neural networks and learning systems*, 33(11):6584–6598, 2021.
- Duan, J., Wang, W., Xiao, L., Gao, J., Li, S. E., Liu, C., Zhang, Y.-Q., Cheng, B., and Li, K. Distributional soft actor-critic with three refinements. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025.
- Eysenbach, B. and Levine, S. Maximum entropy rl (provably) solves some robust rl problems. In *International Conference on Learning Representations*, 2022.
- Fox, R., Pakman, A., and Tishby, N. Taming the noise in reinforcement learning via soft updates. In *Proceedings of the Thirty-Second Conference on Uncertainty in Artificial Intelligence*, pp. 202–211, 2016.
- Fujimoto, S., van Hoof, H., and Meger, D. Addressing function approximation error in actor-critic methods. In *Proceedings of the 35th International Conference on Machine Learning (ICML 2018)*, pp. 1587–1596, Stockholm, Sweden, 2018. PMLR.
- Fujimoto, S., Chang, W.-D., Smith, E., Gu, S. S., Precup, D., and Meger, D. For sale: State-action representation learning for deep reinforcement learning. *Advances in neural information processing systems*, 36:61573–61624, 2023.
- Haarnoja, T., Tang, H., Abbeel, P., and Levine, S. Reinforcement learning with deep energy-based policies. In *International conference on machine learning*, pp. 1352–1361. PMLR, 2017.
- Haarnoja, T., Zhou, A., Abbeel, P., and Levine, S. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *Proceedings of the 35th International Conference on Machine Learning (ICML 2018)*, pp. 1861–1870, Stockholm, Sweden, 2018a. PMLR.
- Haarnoja, T., Zhou, A., Hartikainen, K., Tucker, G., Ha, S., Tan, J., Kumar, V., Zhu, H., Gupta, A., Abbeel, P., et al. Soft actor-critic algorithms and applications. *arXiv preprint arXiv:1812.05905*, 2018b.
- Hao, J., Yang, T., Tang, H., Bai, C., Liu, J., Meng, Z., Liu, P., and Wang, Z. Exploration in deep reinforcement learning: From single-agent to multiagent domain. *IEEE Transactions on Neural Networks and Learning Systems*, 35(7):8762–8782, 2023.

- Hazan, E., Kakade, S., Singh, K., and Van Soest, A. Provably efficient maximum entropy exploration. In *International Conference on Machine Learning*, pp. 2681–2691. PMLR, 2019.
- Hollenstein, J., Auddy, S., Saveriano, M., Renaudo, E., and Piater, J. Action noise in off-policy deep reinforcement learning: Impact on exploration and performance. *arXiv preprint arXiv:2206.03787*, 2022.
- Li, M., Huang, T., and Zhu, W. Adaptive exploration policy for exploration–exploitation tradeoff in continuous action control optimization. *International Journal of Machine Learning and Cybernetics*, 12(12):3491–3501, 2021.
- Li, S. E. *Reinforcement Learning for Sequential Decision and Optimal Control*. Springer Verlag, Singapore, 2023.
- Lillicrap, T. P., Hunt, J. J., Pritzel, A., Heess, N., Erez, T., Tassa, Y., Silver, D., and Wierstra, D. Continuous control with deep reinforcement learning. In *4th International Conference on Learning Representations (ICLR 2016)*, San Juan, Puerto Rico, 2016.
- Nachum, O., Norouzi, M., Xu, K., and Schuurmans, D. Bridging the gap between value and policy based reinforcement learning. In *30th Advances in Neural Information Processing Systems (NeurIPS 2017)*, pp. 2775–2785, Long Beach, CA, USA, 2017.
- Plappert, M., Houthoofd, R., Dhariwal, P., Sidor, S., Chen, R. Y., Chen, X., Asfour, T., Abbeel, P., and Andrychowicz, M. Parameter space noise for exploration. In *International Conference on Learning Representations*, 2018.
- Rawlik, K., Toussaint, M., and Vijayakumar, S. On stochastic optimal control and reinforcement learning by approximate inference. *Proceedings of Robotics: Science and Systems VIII*, 2012.
- Schulman, J., Levine, S., Abbeel, P., Jordan, M. I., and Moritz, P. Trust region policy optimization. In *Proceedings of the 32nd International Conference on Machine Learning (ICML 2015)*, pp. 1889–1897, Lille, France, 2015.
- Schulman, J., Chen, X., and Abbeel, P. Equivalence between policy gradients and soft q-learning. *arXiv preprint arXiv:1704.06440*, 2017a.
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A., and Klimov, O. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017b.
- Seo, Y., Sferrazza, C., Geng, H., Nauman, M., Yin, Z.-H., and Abbeel, P. Fasttd3: Simple, fast, and capable reinforcement learning for humanoid control. *arXiv preprint arXiv:2505.22642*, 2025.
- Sikchi, H., Zhou, W., and Held, D. Learning off-policy with online planning. In *Conference on Robot Learning*, pp. 1622–1633. PMLR, 2022.
- Sun, H., Han, L., Yang, R., Ma, X., Guo, J., and Zhou, B. Exploit reward shifting in value-based deep-rl: Optimistic curiosity-based exploration and conservative exploitation via linear reward shaping. *Advances in neural information processing systems*, 35:37719–37734, 2022.
- Sutton, R. S. and Barto, A. G. *Reinforcement learning: An introduction*. MIT press, 2018.
- Tokic, M. Adaptive ϵ -greedy exploration in reinforcement learning based on value differences. In *Annual conference on artificial intelligence*, pp. 203–210. Springer, 2010.
- Wang, L., Zhang, X., Wang, Y., Zhan, G., Wang, W., Gao, H., Duan, J., and Li, S. E. Off-policy reinforcement learning with model-based exploration augmentation. *Advances in Neural Information Processing Systems*, 38: 92121–92151, 2026.
- Wang, X., Wang, S., Liang, X., Zhao, D., Huang, J., Xu, X., Dai, B., and Miao, Q. Deep reinforcement learning: A survey. *IEEE Transactions on Neural Networks and Learning Systems*, 35(4):5064–5078, 2022.
- Wang, Y., Wang, L., Jiang, Y., Zou, W., Liu, T., Song, X., Wang, W., Xiao, L., Wu, J., Duan, J., et al. Diffusion actor-critic with entropy regulator. *Advances in Neural Information Processing Systems*, 37:54183–54204, 2024.
- Wang, Y., Tan, M., Zou, W., Lin, H., Song, X., Wang, W., Liu, T., Wang, L., Zhan, G., Zhu, T., et al. Enhanced dacer algorithm with high diffusion efficiency. *arXiv preprint arXiv:2505.23426*, 2025.
- Yang, L., Huang, Z., Lei, F., Zhong, Y., Yang, Y., Fang, C., Wen, S., Zhou, B., and Lin, Z. Policy representation via diffusion probability model for reinforcement learning. *arXiv preprint arXiv:2305.13122*, 2023a.
- Yang, L., Zhang, Z., Song, Y., Hong, S., Xu, R., Zhao, Y., Zhang, W., Cui, B., and Yang, M.-H. Diffusion models: A comprehensive survey of methods and applications. *ACM computing surveys*, 56(4):1–39, 2023b.
- Zhan, G., Lyu, Y., Li, S. E., Jiang, Y., Zhang, X., and Tao, L. Enhance generality by model-based reinforcement learning and domain randomization. In *2023 7th CAA International Conference on Vehicular Control and Intelligence (CVCI)*, pp. 1–6. IEEE, 2023.
- Zhan, G., Ge, Q., Gao, H., Yin, Y., Zhao, B., and Eben Li, S. An explicit discrete-time dynamic vehicle model with assured numerical stability. *Vehicle System Dynamics*, pp. 1–24, 2024a.

- Zhan, G., Jiang, Y., Li, S. E., Lyu, Y., Zhang, X., and Yin, Y. A transformation-aggregation framework for state representation of autonomous driving systems. *IEEE Transactions on Intelligent Transportation Systems*, 25 (7):7311–7322, 2024b.
- Zhan, G., Wang, L., Zhang, X., Gao, J., Tomizuka, M., and Li, S. E. Bootstrap off-policy with world model. *Advances in Neural Information Processing Systems*, 38: 134093–134121, 2025a.
- Zhan, G., Zhang, X., Zhang, F., Tao, L., and Li, S. E. Bicriteria policy optimization for high-accuracy reinforcement learning. *IEEE Transactions on Neural Networks and Learning Systems*, 2025b.
- Zhan, G., Tao, L., Wang, P., Wang, Y., Chen, Y., Li, Y., Li, H., Tomizuka, M., and Li, S. Mean flow policy with instantaneous velocity constraint for one-step action generation. In *International Conference on Learning Representations*, volume 2026, pp. 157132–157154, 2026.
- Zhu, Z., Zhao, H., He, H., Zhong, Y., Zhang, S., Guo, H., Chen, T., and Zhang, W. Diffusion models for reinforcement learning: A survey. *arXiv preprint arXiv:2311.01223*, 2023.

A. Theoretical Analysis

A.1. Useful Lemmas

Lemma A.1 (Convergence of γ -Contraction Mappings). *Let (X, d) be a complete metric space, and let $\mathcal{B} : X \rightarrow X$ be a γ -contraction mapping with $0 < \gamma < 1$. This means that for all $x, y \in X$,*

$$d(\mathcal{B}(x), \mathcal{B}(y)) \leq \gamma \cdot d(x, y), \quad (19)$$

where d is the metric on X . According to Banach's fixed-point theorem, $\mathcal{B}$ has a unique fixed point $x^ \in X$, such that $\mathcal{B}(x^*) = x^*$. Furthermore, for any initial point $x_0 \in X$, the iterative sequence $\{x_n\}$ defined by $x_{n+1} = \mathcal{B}(x_n)$ converges to x^* . The convergence rate is geometric, and we have the inequality*

$$d(x_n, x^*) \leq \gamma^n \cdot d(x_0, x^*), \quad \forall n \geq 0. \quad (20)$$

This result not only guarantees the existence and uniqueness of the fixed point but also provides a precise rate at which the sequence approaches x^ , demonstrating the efficiency of contraction mappings in finding fixed points.*

Lemma A.2 (Optimal Policy in Maximum Entropy RL). *For a given state-action value function $Q(s, a)$, the optimal policy $\pi^*(a|s)$ that maximizes the entropy-augmented objective $\mathbb{E}[\sum_{t=0}^{\infty} \gamma^t (r_t + \alpha \mathcal{H}(\pi))]$ is uniquely characterized by the Boltzmann distribution:*

$$\pi^*(a|s) = \frac{1}{Z(s)} \exp\left(\frac{Q(s, a)}{\alpha}\right), \quad (21)$$

where $Z(s) = \int_{\mathcal{A}} \exp(Q(s, a')/\alpha) da'$ is the partition function, and α is the temperature parameter.

A.2. Proof of Trajectory Entropy Bellman Equation

Proof. Here, we build the correspondence between the definition of Q_e in (12) and we proposed entropy Bellman expectation equation in (14).

Q_e defined in (12) represents the cumulative policy entropy starting from the next time step, expressed as:

$$Q_e(s, a) = \mathbb{E}_{\pi} \left[\sum_{t=1}^{\infty} \gamma^t \mathcal{H}(\pi(\cdot|s_t)) \mid s_0 = s, a_0 = a \right]. \quad (22)$$

Our proposed entropy Bellman expectation equation in (14) states

$$Q_e(s, a) = \gamma \mathcal{H}(\pi(\cdot|s')) + \gamma Q_e(s', a'). \quad (23)$$

Substitute the definition of Q_e into the RHS of (14), we have:

$$\begin{aligned} RHS &= \gamma \mathcal{H}(\pi(\cdot|s')) + \gamma Q_e(s', a') \\ &= \gamma \mathcal{H}(\pi(\cdot|s_1)) + \gamma \sum_{t=1}^{\infty} \gamma^t \mathcal{H}(\pi(\cdot|s_{t+1})) \\ &= \gamma \mathcal{H}(\pi(\cdot|s_1)) + \sum_{t=1}^{\infty} \gamma^{t+1} \mathcal{H}(\pi(\cdot|s_{t+1})) \\ &= \gamma \mathcal{H}(\pi(\cdot|s_1)) + \sum_{t=2}^{\infty} \gamma^t \mathcal{H}(\pi(\cdot|s_t)) \\ &= \sum_{t=1}^{\infty} \gamma^t \mathcal{H}(\pi(\cdot|s_t)) = LHS. \end{aligned} \quad (24)$$

Thus, we have proven that the definition of Q_e is the solution of the entropy Bellman expectation equation.

□

A.3. Proof of Convergence of TECRL

Proof. We prove the convergence of PIS by showing that the entropy Bellman operator $\mathcal{B}_e$, defined as

$$\mathcal{B}_e Q_e(s, a) = \gamma[Q_e(s', a') - \log \pi(a'|s')], \quad (25)$$

is a γ -contraction mapping.

We analyze the infinity norm of $\mathcal{B}_e$. For any two functions $Q_{e,1}(s, a)$ and $Q_{e,2}(s, a)$, we have:

$$\begin{aligned} \|\mathcal{B}_e[Q_{e,1}(s, a)] - \mathcal{B}_e[Q_{e,2}(s, a)]\|_\infty &= \|\gamma[Q_{e,1}(s', a') - \log \pi(a'|s')] \\ &\quad - \gamma[Q_{e,2}(s', a') - \log \pi(a'|s')]\|_\infty \\ &\leq \|\gamma Q_{e,1}(s', a') - \gamma Q_{e,2}(s', a')\|_\infty \\ &= \gamma\|Q_{e,1}(s', a') - Q_{e,2}(s', a')\|_\infty. \end{aligned} \quad (26)$$

Since $\gamma \in (0, 1)$, it follows that $\mathcal{B}_e$ is a γ -contraction mapping. By applying Lemma A.1, we know that $\mathcal{B}_e$ has a unique fixed point. This fixed point can be obtained by iteratively applying $\mathcal{B}_e$ starting from an arbitrary initial $Q_{e,\text{init}}$. That is, as the iteration number k increases, the sequence of updated Q functions converges to a fixed point, i.e., the desired Q_e .

This completes the proof. $\square$

A.4. Proof of Optimal Policy Form of TECRL

Proof. In maximum entropy RL, the Boltzmann distribution of value function is as follows: $p^*(a|s) = \frac{1}{Z(s)} \exp\left(\frac{Q(s,a)}{\alpha}\right)$. The core idea of TECRL is to introduce a constraint that the trajectory entropy equals a predefined entropy budget $\mathcal{H}_{\text{budget}}$. So the problem can be expressed as:

$$\begin{aligned} \max_{\pi} \quad & \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t [r(s_t, a_t)] \right] \\ \text{s.t.} \quad & \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t \mathcal{H}(\pi(\cdot|s_t)) \right] = \mathcal{H}_{\text{budget}}. \end{aligned} \quad (27)$$

In fact, introducing a trajectory-entropy constraint is essentially equivalent to using a weight function $w(s, a) \propto \exp(Q_e(s, a))$ to refine the original Boltzmann distribution:

$$\hat{p}(a | s) \propto \underbrace{\exp\left(\frac{1}{\alpha} Q_r(s, a)\right)}_{\text{reward Boltzmann}} \cdot \underbrace{w(s, a)}_{\text{TEC-refinement}} \quad (28)$$

To align the policy π with $\hat{p}$, we minimize their Kullback-Leibler (KL) divergence:

$$\begin{aligned} \mathbb{D}_{\text{KL}}(\pi || \hat{p}) &= \int \pi(a|s) \log \frac{\pi(a|s)}{\hat{p}(a|s)} da \\ &= \int \pi(a|s) \log \pi(a|s) da - \int \pi(a|s) \log \hat{p}(a|s) da \\ &= -\mathcal{H}(\pi(\cdot|s)) - \mathbb{E}_{a \sim \pi} [\log \hat{p}(a|s)] \\ &= -\mathcal{H}(\pi(\cdot|s)) - \mathbb{E}_{a \sim \pi} \left[\frac{1}{\alpha} Q_r(s, a) + Q_e(s, a) - \log Z \right] \\ &= -\frac{1}{\alpha} \mathbb{E}_{\pi} [Q_r(s, a) + \alpha Q_e(s, a) + \alpha \mathcal{H}(\pi(\cdot|s))] + \log Z(s). \end{aligned}$$

Minimizing this divergence is equivalent to maximizing the policy objective $J_{\pi} = Q_r + \alpha Q_e + \alpha \mathcal{H}(\pi)$. The minimum is achieved when $\pi = \hat{p}$, yielding the closed-form solution.

This completes the proof. $\square$

A.5. Proof of Performance Bound of TECRL

Proof. By definition, π_{soft}^* maximizes the soft objective $J_{\text{MaxEnt}}(\pi) = R(\pi) + \alpha_{\text{soft}}^* \mathcal{H}(\pi)$. Therefore, for any policy π (including the optimal TECRL policy π_{TEC}^*), we have:

$$R_{\text{MaxEnt}}^* + \alpha_{\text{soft}}^* \mathcal{H}_{\text{soft}}^* \geq R(\pi) + \alpha_{\text{soft}}^* \mathcal{H}(\pi). \quad (29)$$

Under the TECRL constraint, $\mathcal{H}(\pi_{\text{TEC}}^*) = \mathcal{H}_{\text{budget}}$. Substituting this into the inequality:

$$\begin{aligned} R_{\text{MaxEnt}}^* + \alpha_{\text{soft}}^* \mathcal{H}_{\text{soft}}^* &\geq R_{\text{TEC}}^* + \alpha_{\text{soft}}^* \mathcal{H}(\pi_{\text{TEC}}^*) \\ &= R_{\text{TEC}}^* + \alpha_{\text{soft}}^* \mathcal{H}_{\text{budget}}. \end{aligned} \quad (30)$$

Rearranging the terms to isolate R_{TEC}^* :

$$R_{\text{TEC}}^* \leq R_{\text{MaxEnt}}^* + \alpha_{\text{soft}}^* (\mathcal{H}_{\text{soft}}^* - \mathcal{H}_{\text{budget}}). \quad (31)$$

Thus, we have

$$\Delta = R_{\text{TEC}}^* - R_{\text{MaxEnt}}^* \leq \alpha_{\text{soft}}^* (\mathcal{H}_{\text{soft}}^* - \mathcal{H}_{\text{budget}}). \quad (32)$$

This completes the proof. $\square$

B. Evidence of mitigating identified two bottlenecks

B.1. Non-stationary of Q-value target

We acknowledge that the bootstrapping update mechanism of Q-values contributes significantly to the non-stationarity in RL. In this context, we highlight that the coupling of reward and entropy is another crucial contributing factor, and our method can effectively address and eliminate this factor. As shown in Figure 7, we provide direct empirical evidence via Q-loss and Q-std curves (comparing DSAC-T and our DSAC-E) on the Humanoid task:

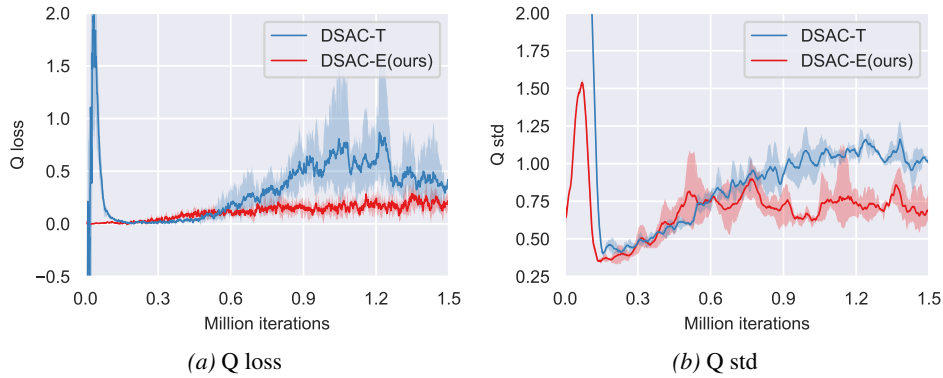


Figure 7. Evidence of mitigating (1) non-stationary Q-value estimation.

Q-loss:

- DSAC-T exhibits a sharp initial peak and substantial late-stage oscillations.
- In contrast, DSAC-E maintains stable Q-loss throughout training. Its slight late-stage increase mainly aligns with the growing magnitude of Q-values as task performance improves, rather than instability.

Q-std

- DSAC-T shows a significantly higher peak at the early stage.
- During the mid-to-late training phases, our DSAC-E’s Q-std remains consistently lower than that of DSAC-T, indicating more stable Q-value estimates.

We emphasize that the observed differences in Q-loss and Q-std between DSAC-T and DSAC-E are mainly attributed to the RES’s elimination of alpha-related non-stationarity, while all other components (policy update, NN structure, etc.) are kept consistent. This controlled comparison provides a direct evidence for non-stationary Q-value and directly links the improved stability to our proposed RES.

B.2. Short-sighted local tuning

We acknowledge that maximum entropy RL, such as SAC and DSAC-T, have achieved notable success with their local entropy tuning, but our experiments reveal this mechanism can be improved by equipping the entropy adaptation capability that our TEC aims to provide. To substantiate this and validate the “strategic distribution” of entropy, we conducted targeted analyzes on the Humanoid and Walker2d tasks.

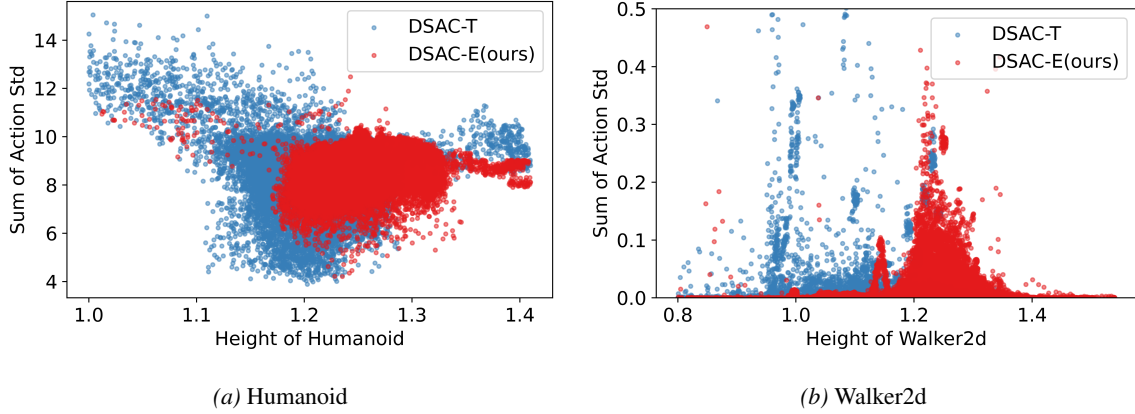


Figure 8. Evidence for short-sighted local entropy tuning.

We evaluated 50 episodes per task for both DSAC-T and our DSAC-E, then visualized the sum of action std, proportional to the policy entropy, across the state space. As shown in Figure 8, for intuitive presentation, we set:

- X-axis: Robot’s center-of-mass (CoM) height (a key, interpretable state variable, and extreme values (too low/high) trigger termination as dangerous states).
- Y-axis: Sum of action standard deviations, which positively correlates with entropy for diagonal Gaussian policies.

From these two visualizations of entropy distribution, one can observe that:

- DSAC-E consistently exhibits higher entropy in the middle CoM height range (normal states) and lower entropy at side extreme CoM heights (dangerous states).
- DSAC-T shows no such state-aware adaptation, its entropy distribution is relatively uniform (compare to ours) across the entire CoM height range, failing to adjust to state-specific risks.

This confirms that TEC does not merely optimize average entropy. Instead, it enables strategic entropy allocation: preserving exploration in normal states while reducing randomness in high-risk regions to avoid termination. This aligns with the core motivation of addressing local tuning’s short-sightedness. Local tuning (as in DSAC-T) treats all states equally, whereas TEC restrict the policy entropy at the trajectory level, allowing adaptive exploration allocation.

C. Error Propagation Analysis of TECRL

This appendix analyzes the coupled learning dynamics of the dual-critic architecture in TECRL. In particular, we investigate how critic estimation errors propagate through policy improvement, temperature updating, and subsequent Bellman target construction. We further compare this propagation mechanism with that of standard Soft Actor-Critic (SAC), highlighting the structural role of the dual-critic decomposition in suppressing cascading error amplification.

For brevity, the superscript π of the value functions is omitted throughout this section.

C.1. Preliminaries and Error Assumptions

TECRL separately maintains a reward critic Q_r and an entropy critic Q_e . Let Q_r^* and Q_e^* denote their ideal values, and let $\hat{Q}_r$ and $\hat{Q}_e$ denote the corresponding approximations. We assume that the critic estimation errors are uniformly bounded as

$$\|\hat{Q}_r - Q_r^*\|_\infty \leq \epsilon_r, \quad \|\hat{Q}_e - Q_e^*\|_\infty \leq \epsilon_e. \quad (33)$$

We further assume bounded critic values:

$$\max \left\{ \|\hat{Q}_r\|_\infty, \|Q_r^*\|_\infty \right\} \leq Q_{r,\max}, \quad \max \left\{ \|\hat{Q}_e\|_\infty, \|Q_e^*\|_\infty \right\} \leq Q_{e,\max}. \quad (34)$$

For a fixed temperature parameter α , the ideal policy-improvement target of TECRL is given by

$$\pi^*(a|s) = \frac{1}{Z(s)} \exp \left(\frac{Q_r^*(s, a)}{\alpha} + Q_e^*(s, a) \right), \quad (35)$$

where $Z(s)$ denotes the normalization factor.

Using the approximate critics, the corresponding perturbed policy target is

$$\hat{\pi}(a|s) = \frac{1}{\hat{Z}(s)} \exp \left(\frac{\hat{Q}_r(s, a)}{\alpha} + \hat{Q}_e(s, a) \right). \quad (36)$$

The induced perturbation of the policy logits is therefore bounded by

$$\epsilon_{\text{tot}} \triangleq \frac{\epsilon_r}{\alpha} + \epsilon_e. \quad (37)$$

In the following, we trace how this initial critic error propagates through the major learning components of TECRL.

C.2. Error Propagation from Critics to the Policy

We first characterize how critic estimation errors perturb the policy improvement target.

Proposition 1 (Bounded policy deviation). Suppose that the critic errors satisfy Eq. (33). Then the density ratio between the perturbed policy $\hat{\pi}$ and the ideal policy π^* satisfies

$$\exp(-2\epsilon_{\text{tot}}) \leq \frac{\hat{\pi}(a|s)}{\pi^*(a|s)} \leq \exp(2\epsilon_{\text{tot}}). \quad (38)$$

Consequently, their total variation (TV) divergence is bounded by

$$D_{\text{TV}}(\hat{\pi}, \pi^*) \leq 1 - \exp(-2\epsilon_{\text{tot}}) \leq 2\epsilon_{\text{tot}}. \quad (39)$$

Proof. Define the logit perturbation as

$$\delta(s, a) \triangleq \frac{\hat{Q}_r(s, a) - Q_r^*(s, a)}{\alpha} + \hat{Q}_e(s, a) - Q_e^*(s, a). \quad (40)$$

According to Eq. (33),

$$|\delta(s, a)| \leq \epsilon_{\text{tot}}. \quad (41)$$

The corresponding normalization factors satisfy

$$\exp(-\epsilon_{\text{tot}}) \leq \frac{\hat{Z}(s)}{Z(s)} \leq \exp(\epsilon_{\text{tot}}). \quad (42)$$

Combining Eqs. (41) and (42) yields

$$\exp(-2\epsilon_{\text{tot}}) \leq \frac{\hat{\pi}(a|s)}{\pi^*(a|s)} \leq \exp(2\epsilon_{\text{tot}}). \quad (43)$$

Using the definition

$$D_{\text{TV}}(\hat{\pi}, \pi^*) = \int_{\pi^* > \hat{\pi}} (\pi^*(a|s) - \hat{\pi}(a|s)) da, \quad (44)$$

we obtain

$$D_{\text{TV}}(\hat{\pi}, \pi^*) \leq 1 - \exp(-2\epsilon_{\text{tot}}). \quad (45)$$

Finally, using $1 - e^{-x} \leq x$ for $x \geq 0$ gives Eq. (39). $\square$

Proposition 1 shows that bounded critic errors lead to a bounded policy distribution shift:

$$D_{\text{TV}}(\hat{\pi}, \pi^*) = \mathcal{O}\left(\frac{\epsilon_r}{\alpha} + \epsilon_e\right). \quad (46)$$

Thus, sufficiently small critic estimation errors induce only a first-order perturbation of the policy-improvement target.

To isolate the effect of the entropy critic in the subsequent analysis, we consider the special case in which the reward critic is fixed and only Q_e is perturbed. In this case, $\epsilon_{\text{tot}} = \epsilon_e$, and

$$D_{\text{TV}}(\hat{\pi}, \pi^*) \leq 1 - \exp(-2\epsilon_e) \leq 2\epsilon_e. \quad (47)$$

C.3. Error Propagation to the Temperature Update

The policy perturbation subsequently influences the temperature update. Let g denote the ideal temperature gradient and $\hat{g}$ denote the gradient evaluated using the approximate entropy critic and the perturbed policy.

We focus on the entropy-critic perturbation because it directly captures the coupling among entropy estimation, policy improvement, and temperature adaptation.

Proposition 2 (Bounded temperature-update error). Suppose that

$$\left\| \hat{Q}_e - Q_e^* \right\|_{\infty} \leq \epsilon_e. \quad (48)$$

For a temperature learning rate η_{α} , the error introduced into one temperature update satisfies

$$|\Delta\alpha_{\text{err}}| = \eta_{\alpha} |\hat{g} - g| \leq 3\eta_{\alpha}\epsilon_e. \quad (49)$$

Proof. The perturbation of the temperature gradient can be decomposed into the contributions from the policy log-density and the entropy critic:

$$|\hat{g} - g| \leq \|\log \hat{\pi} - \log \pi^*\|_{\infty} + \left\| \hat{Q}_e - Q_e^* \right\|_{\infty}. \quad (50)$$

According to the density-ratio bound in Eq. (38),

$$\|\log \hat{\pi} - \log \pi^*\|_{\infty} \leq 2\epsilon_e. \quad (51)$$

Therefore,

$$|\hat{g} - g| \leq 2\epsilon_e + \epsilon_e = 3\epsilon_e, \quad (52)$$

which directly yields Eq. (49). $\square$

Proposition 2 shows that critic estimation errors can generate transient temperature fluctuations. The key question is whether such temperature errors are subsequently injected back into critic learning through the Bellman targets. This issue is examined next.

C.4. Feedback to the Bellman Targets

We now analyze how the policy deviation induced by the critic error feeds back into the Bellman targets in the subsequent policy-evaluation step.

The reward critic target of TECRL is

$$y_r(s, a) = r(s, a) + \gamma \mathbb{E}_{a' \sim \pi(\cdot|s')} [Q_r(s', a')], \quad (53)$$

while the entropy critic target is

$$y_e(s, a) = \gamma \mathbb{E}_{a' \sim \pi(\cdot|s')} [-\log \pi(a'|s') + Q_e(s', a')]. \quad (54)$$

A key structural property of Eqs. (53)–(54) is that neither Bellman target explicitly contains the temperature parameter α .

Proposition 3 (Bounded Bellman-target perturbation). Consider the entropy-critic perturbation

$$\|\hat{Q}_e - Q_e^*\|_\infty \leq \epsilon_e, \quad (55)$$

with the reward critic fixed. Then the induced perturbations of the reward and entropy Bellman targets satisfy

$$|\hat{y}_r - y_r^*| \leq 4\gamma Q_{r,\max} \epsilon_e, \quad (56)$$

$$|\hat{y}_e - y_e^*| \leq \gamma (3 + 4Q_{e,\max}) \epsilon_e. \quad (57)$$

Moreover, when the policy is treated as fixed during Bellman target evaluation,

$$\frac{\partial y_r}{\partial \alpha} = 0, \quad \frac{\partial y_e}{\partial \alpha} = 0. \quad (58)$$

Hence, the temperature-update error $\Delta\alpha_{\text{err}}$ does not enter either critic target directly.

Proof. For the reward target, the perturbation induced by the policy distribution shift is bounded by

$$\begin{aligned} |\hat{y}_r - y_r^*| &\leq \gamma |\mathbb{E}_{\hat{\pi}}[Q_r^*] - \mathbb{E}_{\pi^*}[Q_r^*]| \\ &\leq \gamma \|Q_r^*\|_\infty \|\hat{\pi} - \pi^*\|_1 \\ &= 2\gamma Q_{r,\max} D_{\text{TV}}(\hat{\pi}, \pi^*). \end{aligned} \quad (59)$$

Using Eq. (47), we obtain

$$|\hat{y}_r - y_r^*| \leq 4\gamma Q_{r,\max} \epsilon_e. \quad (60)$$

For the entropy target, the perturbation can be decomposed into three components: the log-policy shift, the entropy-critic estimation error, and the policy-distribution shift. Accordingly,

$$\begin{aligned} |\hat{y}_e - y_e^*| &\leq \gamma \|\log \hat{\pi} - \log \pi^*\|_\infty + \gamma \|\hat{Q}_e - Q_e^*\|_\infty \\ &\quad + \gamma \|Q_e^*\|_\infty \|\hat{\pi} - \pi^*\|_1. \end{aligned} \quad (61)$$

Applying Eqs. (47) and (51) yields

$$\begin{aligned} |\hat{y}_e - y_e^*| &\leq 2\gamma\epsilon_e + \gamma\epsilon_e + 4\gamma Q_{e,\max}\epsilon_e \\ &= \gamma(3 + 4Q_{e,\max})\epsilon_e. \end{aligned} \quad (62)$$

Finally, because α does not explicitly appear in either Eq. (53) or Eq. (54), their direct sensitivities to the temperature parameter vanish when the policy is held fixed, yielding Eq. (58). $\square$

Proposition 3 shows that the critic-to-policy-to-critic feedback remains linearly bounded by the initial critic estimation error. More importantly, although an entropy-critic error can perturb the temperature update, the resulting temperature error is not directly injected into either Bellman target.

C.5. Comparison with the Coupled Target in SAC

The error-isolation property of TECRL differs fundamentally from the target construction in standard Soft Actor-Critic (SAC). In SAC, reward and entropy regularization are combined within a single Bellman target:

$$y_{\text{SAC}} = r + \gamma \mathbb{E}_{a' \sim \pi(\cdot|s')} [Q(s', a') - \alpha \log \pi(a'|s')]. \quad (63)$$

Let the estimated temperature be written as

$$\hat{\alpha} = \alpha^* + \Delta\alpha_{\text{err}}, \quad (64)$$

where α^* denotes the ideal temperature and $\Delta\alpha_{\text{err}}$ denotes the temperature estimation error.

Because α explicitly appears in the SAC Bellman target, the target perturbation contains a direct temperature-error term. Schematically, the resulting target shift can be decomposed as

$$\begin{aligned} |\Delta y_{\text{SAC}}| &\lesssim \underbrace{\gamma\epsilon_Q}_{\text{critic error}} + \underbrace{2\gamma Q_{\max} D_{\text{TV}}(\hat{\pi}, \pi^*)}_{\text{policy distribution shift}} \\ &\quad + \underbrace{\gamma |\Delta\alpha_{\text{err}}| \|\log \pi^*\|_{\infty}}_{\text{direct temperature-error injection}} \\ &\quad + \underbrace{\gamma\alpha^* \|\log \hat{\pi} - \log \pi^*\|_{\infty}}_{\text{entropy-term perturbation}}. \end{aligned} \quad (65)$$

Hence, a critic estimation error in SAC can propagate through the following feedback pathway:

$$\boxed{\epsilon_Q \rightarrow \Delta\pi \rightarrow \Delta\alpha \rightarrow \Delta y_Q \rightarrow \epsilon_Q^+}, \quad (66)$$

where ϵ_Q^+ denotes the critic error induced in the subsequent iteration. The explicit dependence of the critic target on α thus introduces an additional channel through which temperature fluctuations can feed back into critic learning.

In contrast, TECRL separates reward evaluation and entropy evaluation into two Bellman targets:

$$y_r = r + \gamma \mathbb{E}_{a' \sim \pi(\cdot|s')} [Q_r(s', a')], \quad (67)$$

$$y_e = \gamma \mathbb{E}_{a' \sim \pi(\cdot|s')} [-\log \pi(a'|s') + Q_e(s', a')]. \quad (68)$$

The temperature parameter does not explicitly appear in either target. Accordingly,

$$\left. \frac{\partial y_r}{\partial \alpha} \right|_{\pi} = 0, \quad \left. \frac{\partial y_e}{\partial \alpha} \right|_{\pi} = 0, \quad (69)$$

where the subscript π emphasizes that the policy is held fixed during Bellman evaluation.

The corresponding target perturbations can therefore be summarized as

$$|\Delta y_r| \lesssim \underbrace{\gamma \epsilon_r}_{\text{reward-critic error}} + \underbrace{2\gamma Q_{r,\max} D_{\text{TV}}(\hat{\pi}, \pi^*)}_{\text{policy distribution shift}}, \quad (70)$$

$$\begin{aligned} |\Delta y_e| \lesssim & \underbrace{\gamma \epsilon_e}_{\text{entropy-critic error}} + \underbrace{2\gamma Q_{e,\max} D_{\text{TV}}(\hat{\pi}, \pi^*)}_{\text{policy distribution shift}} \\ & + \underbrace{\gamma \|\log \hat{\pi} - \log \pi^*\|_{\infty}}_{\text{log-policy perturbation}}. \end{aligned} \quad (71)$$

Notably, neither Eq. (70) nor Eq. (71) contains a term proportional to $|\Delta \alpha_{\text{err}}|$.

Therefore, although critic errors can still perturb the policy and subsequently affect Bellman evaluation through the induced distribution shift, temperature fluctuations do not constitute an additional *direct* disturbance to the critic targets in TECRL.

The distinction between the two error-propagation structures can be summarized as

$$\text{SAC:} \quad \epsilon_Q \rightarrow \Delta \pi \rightarrow \Delta \alpha \rightarrow \Delta y_Q, \quad (72)$$

$$\text{TECRL:} \quad (\epsilon_r, \epsilon_e) \rightarrow \Delta \pi \rightarrow (\Delta y_r, \Delta y_e), \quad \Delta \alpha \not\rightarrow (\Delta y_r, \Delta y_e) \text{ directly.} \quad (73)$$

Thus, the dual-critic decomposition does not eliminate all coupling among the learning components. Instead, it specifically removes the direct temperature-to-critic feedback channel induced by the coupled SAC target.

C.6. Discussion of the Coupled Learning Dynamics

The complete error-propagation mechanism of TECRL can now be summarized in two interacting pathways.

First, critic estimation errors perturb the policy-improvement target and subsequently the temperature update:

$$(\epsilon_r, \epsilon_e) \rightarrow D_{\text{TV}}(\hat{\pi}, \pi^*) \rightarrow \Delta \alpha_{\text{err}}. \quad (74)$$

Second, the induced policy distribution shift feeds back into the Bellman targets:

$$(\epsilon_r, \epsilon_e) \rightarrow D_{\text{TV}}(\hat{\pi}, \pi^*) \rightarrow (\Delta y_r, \Delta y_e). \quad (75)$$

Crucially, the direct pathway

$$\Delta \alpha_{\text{err}} \not\rightarrow (\Delta y_r, \Delta y_e) \quad \text{during Bellman evaluation} \quad (76)$$

is absent from TECRL.

This structural decoupling does not imply that TECRL is completely insensitive to critic approximation errors. Critic errors still induce a policy distribution shift, and the perturbed policy subsequently changes the state-action distribution used for policy evaluation. Nevertheless, Propositions 1–3 show that, under bounded critic errors, these perturbations remain first-order bounded.

Specifically,

$$D_{\text{TV}}(\hat{\pi}, \pi^*) = \mathcal{O}\left(\frac{\epsilon_r}{\alpha} + \epsilon_e\right). \quad (77)$$

When focusing on the entropy-critic perturbation, the propagated errors satisfy

$$|\Delta y_r|, \quad |\Delta y_e|, \quad |\Delta \alpha_{\text{err}}| = \mathcal{O}(\epsilon_e). \quad (78)$$

By comparison, the SAC Bellman target contains an additional perturbation term proportional to

$$|\Delta \alpha_{\text{err}}| \|\log \pi^*\|_{\infty}, \quad (79)$$

which directly couples temperature fluctuations back into critic learning.

Therefore, the primary stability advantage of the TECRL decomposition is not the complete elimination of learning coupling, but the removal of this direct temperature-to-critic feedback channel. The remaining error propagation is primarily mediated by bounded critic approximation errors and the corresponding policy-distribution shift. This structural isolation reduces one potential mechanism of cascading error amplification and thereby improves the robustness of policy evaluation against fluctuations in the temperature update.

D. Environmental Introduction

In this paper, we use 10 widely adopted tasks, covering locomotion, manipulation and visual driving domains: Humanoid-v3, Ant-v3, HalfCheetah-v3, Walker2d-v3, InvertedDoublePendulum-v3 (InvertedDP-v2), Hopper-v3, Reacher-v2, Swimmer-v3, Pusher-v2 and Carracing-v2, as illustrated in Figure 9. The InvertedDP-v3 task entails maintaining the balance of a double pendulum in an inverted state. In contrast, the objective of the other tasks is to maximize the forward velocity while avoiding falling. All these tasks are realized through the OpenAI Gym interface (Brockman et al., 2016).

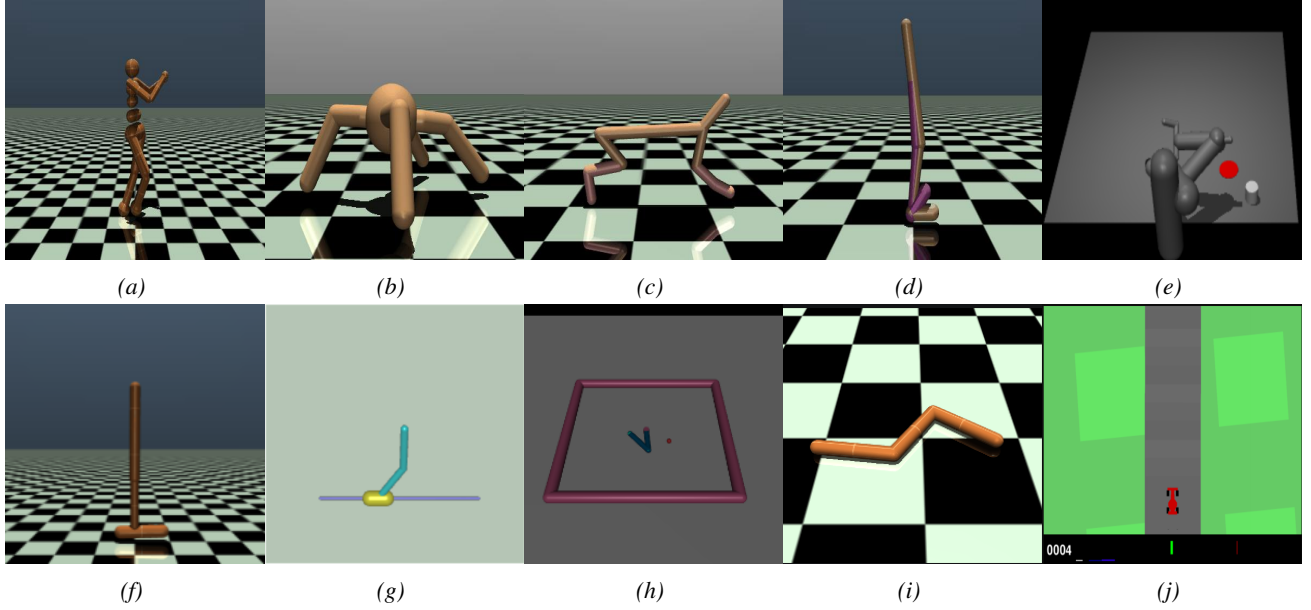


Figure 9. Benchmarks. (a) Humanoid-v3: $(s \times a) \in \mathbb{R}^{376} \times \mathbb{R}^{17}$. (b) Ant-v3: $(s \times a) \in \mathbb{R}^{111} \times \mathbb{R}^8$. (c) HalfCheetah-v3: $(s \times a) \in \mathbb{R}^{17} \times \mathbb{R}^6$. (d) Walker2d-v3: $(s \times a) \in \mathbb{R}^{17} \times \mathbb{R}^6$. (e) Pusher-v2: $(s \times a) \in \mathbb{R}^{23} \times \mathbb{R}^7$. (f) Hopper-v3: $(s \times a) \in \mathbb{R}^{11} \times \mathbb{R}^3$. (g) InvertedDoublePendulum-v2: $(s \times a) \in \mathbb{R}^6 \times \mathbb{R}^1$. (h) Reacher-v2: $(s \times a) \in \mathbb{R}^{11} \times \mathbb{R}^2$. (j) Swimmer-v3: $(s \times a) \in \mathbb{R}^8 \times \mathbb{R}^2$. (i) CarRacing-v1: $(s \times a) \in \mathbb{R}^{96 \times 96 \times 3} \times \mathbb{R}^2$ (image-input).

E. Time-efficiency of Training and Inference

For the sake of training efficiency, we employ three tasks spanning low, medium, and high dimensions (Hopper, Walker, and Humanoid) to compare two algorithms: DSAC-T and our proposed DSAC-E. All experiments are conducted on a single NVIDIA RTX 3090 Ti GPU paired with an AMD Ryzen Threadripper 3960X 24-Core Processor, using the Jax 0.4.28 programming framework.

Regarding inference time-efficiency, our DSAC-E trains a same-size MLP policy as DSAC-T, so the inference time should be identical in principle.

Algorithm	Humanoid (376,17)	Ant (111,8)	Walker2d (17,6)
DSAC-T	2h09m (129)	1h47m (107)	1h29m (89)
DSAC-E	2h23m (143)	2h02m (122)	1h44m (104)
Percentage	110.8%	114.0%	116.9%

F. Reproducibility Statement

The hyperparameters of all algorithms are shown in Table 4. Our proposed DSAC-E algorithm is built on the DSAC-T, inheriting all of its hyperparameters. For the newly introduced hyperparameter ρ , we set its default value as 20.

Table 4
DETAILED HYPERPARAMETERS.

Hyperparameters	Value
<i>Shared</i>	
Optimizer	Adam ($\beta_1 = 0.9, \beta_2 = 0.999$)
Actor learning rate	$1e-4$
Critic learning rate	$1e-4$
Discount factor (γ)	0.99
Policy update interval	2
Target smoothing coefficient (τ)	0.005
Reward scale	0.1
Number of iterations	1.5×10^6
<i>Maximum-entropy framework</i>	
Learning rate of temperature α	3×10^{-4}
Base expected entropy ($\bar{\mathcal{H}}$)	$\bar{\mathcal{H}} = -\dim(\mathcal{A})$
<i>Deterministic policy</i>	
Exploration noise	$\epsilon \sim \mathcal{N}(0, 0.1^2)$
<i>Off-policy</i>	
Sample batch size	20
Replay batch size	256
Replay buffer warm size	1×10^4
Replay buffer size	1×10^6
<i>On-policy</i>	
Sample batch size	2000
Replay batch size	2000
GAE factor	0.95
<i>Distributional Q and budget</i>	
Variance clipping constant ζ	3
Stabilizing constant ϵ and ϵ_ω	0.1
Budget coefficient ρ	20

F.1. Comparison of DSAC-T and DSAC-E with varying trajectory entropy budget

We apply different ρ values to allow varying trajectory entropy budgets for both DSAC-T and our DSAC-E. As shown in Figure 10, the performance gain of DSAC-T (Figure 10a) is not significant with the adjustment of ρ . Its performance varies only slightly and all results cluster closely together. In contrast, our DSAC-E (Figure 10b) consistently outperforms DSAC-T across all settings, and its performance shows a clearer, more structured dependence on ρ .

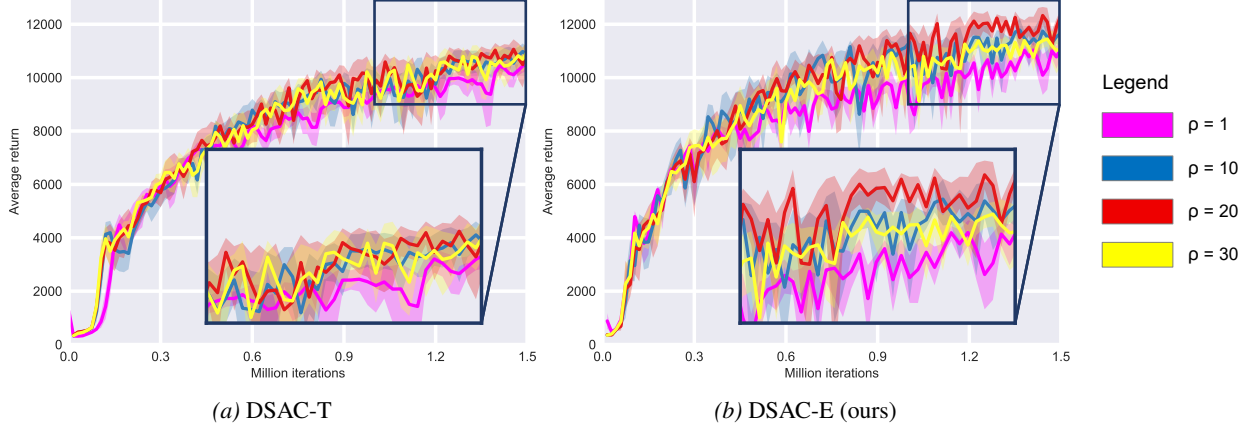


Figure 10. Ablation on the sensitivity to the trajectory entropy budget.

For both DSAC-T and our DSAC-E, performance first improves and then degrades as ρ increases, which aligns with our theoretical analysis: a properly chosen entropy budget can lift the performance bound, whereas an excessively large ρ (corresponding to an overly small entropy budget) reduces exploration and leads to a performance drop. Overall, our DSAC-E achieves higher performance and exhibits a more interpretable sensitivity to ρ , making it easier to tune for high returns.

Table 5. Performance of DSAC-T and our DSAC-E under different ρ values.

Algorithm	$\rho = 1$	$\rho = 10$	$\rho = 20$	$\rho = 30$
DSAC-T	10829 ± 243	11079 ± 457	11455 ± 404	11182 ± 705
DSAC-E (ours)	11382 ± 447	12118 ± 505	12542 ± 280	11747 ± 365

* Note the base entropy budget is $-\dim(\mathcal{A})$, so higher ρ means lower entropy budget $-\rho \dim(\mathcal{A})$.

G. Visualizations

To demonstrate the effectiveness of DSAC-E in solving complex, high-dimensional locomotion tasks, we provide visualizations of policy control process on two most challenging tasks

G.1. Humanoid task

The visualization of Humanoid task in Figure 11 showcase that DSAC-E not only achieves successful locomotion but also develops a proactive gait, characterized by a **forward-leaning torso** and **vigorous leg drive**, highlighting its superior capability in managing high-dimensional, challenging control scenarios.



Figure 11. Visualizations of control processes on Humanoid-v3 task.

G.2. Ant task

The visualization of Ant task in Figure 12 showcase that DSAC-E learns coordinated **four-leg locomotion** and achieves better performance, whereas baselines such as DSAC-T typically converge to suboptimal **three-leg locomotion**.

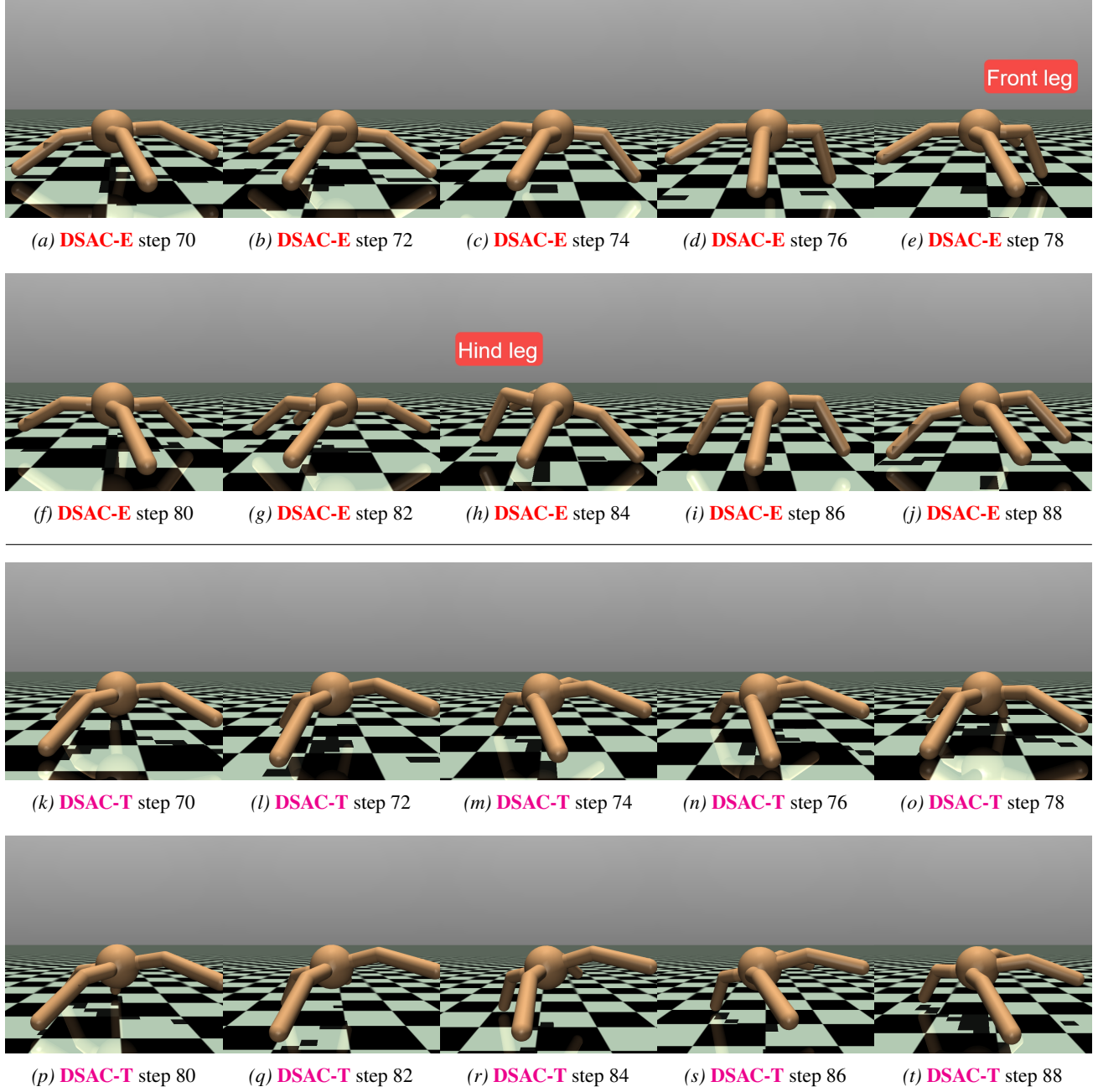


Figure 12. Visualizations of control processes on Ant task.